BM4Implicit with Hairer's Symmetric Projection

Reduced Newton formulation for the projection multiplier

One symmetric projection around a complete twelve-stage BM4 cycle

Scope of the retained coupled derivation

This derivation retains the historical generalized map with harmonic coupling. Current BM4Implicit and BM4Midpoint use its uncoupled specialization $\omega = 0$ and omit C entirely in execution. Coupled formulas and the recorded $\omega = \pi/8$ experiment below describe the earlier configuration, not the current public constructors. The independent C operation remains available as `HarmonicCoupling`; see `theory.tex` for the current entry point.

1 Scope

The method studied here is denoted `BM4Implicit`. It applies Hairer's symmetric projection once around the complete fourth-order BM4 map. All twelve BM4 stages are traversed without an intermediate projection and one multiplier is determined for the complete cycle.

The nonlinear unknown is the physical multiplier only. For one guiding-centre particle it has two components and Newton solves a 2×2 system. For N particles the current implementation assembles a dense $2N \times 2N$ matrix, although the field and coupling maps act independently on each particle.

2 Physical and doubled guiding-centre states

The physical state is $z = (X, Y) \in \mathbb{R}^2$ and satisfies

$$\dot{z} = f(z,t), \qquad f(z,t) = J_0 \nabla_z \phi(z,t), \qquad J_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

The doubled state is

$$\mathcal{Y} = (u, v) \in \mathbb{R}^4, \qquad \mathcal{M} = \{(u, v) : u = v\}.$$

Introduce the diagonal embedding, constraint, and normal embedding

$$E = \begin{pmatrix} I_2 \\ I_2 \end{pmatrix}, \qquad G = (I_2 \ -I_2), \qquad N = G^T = \begin{pmatrix} I_2 \\ -I_2 \end{pmatrix}.$$

They satisfy

$$GE = 0, \qquad GN = 2I_2.$$

Thus $Ez = (z, z)$ is physical and $N\mu = (\mu, -\mu)$ separates the two copies in opposite directions.

3 Direct, adjoint, and coupling maps

For a duration s and evaluation time t, the uncoupled triangular shears are

$$B_{s,t}(u, v) = (u, v + sf(u,t)), \qquad A_{s,t}(u, v) = (u + sf(v,t), v).$$

The GC extended formulation also contains an exact linear coupling $C_{s,\omega}$. If

$$m = \frac{u+v}{2}, \qquad \delta = u - v,$$

then

$$m^+ = m, \qquad \delta^+ = R(2\omega s)\delta, \qquad R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}.$$

The direct map implemented by `GCExtendedFormulation` is

$$D_{s,t} = C_{s,\omega} \circ A_{s,t} \circ B_{s,t},$$

whereas its adjoint is

$$D^*_{s,t} = B_{s,t} \circ A_{s,t} \circ C_{s,\omega}.$$

The ordering is important: the coupling follows both triangular updates in the direct map and precedes them in the adjoint map.

Let $W(z,t) = D_z f(z,t)$. Before the final coupling, the direct-map Jacobian is

$$\begin{pmatrix} I_2 + s^2 W(v^+,t)W(u,t) & sW(v^+,t) \\ sW(u,t) & I_2 \end{pmatrix}, \qquad v^+ = v + sf(u,t).$$

Therefore its full Jacobian is the coupling matrix multiplied on the left by this block matrix. The adjoint Jacobian has the analogous reverse ordering, with the coupling matrix multiplied on the right.

4 The twelve-stage BM4 cycle

The six half-sequence coefficients are

$$\begin{aligned} a_1 &= 0.0792036964311957, & a_2 &= 0.1303114101821663, \\ a_3 &= 0.2228614958676077, & a_4 &= -0.3667132690474257, \\ a_5 &= 0.3246481886897062, & a_6 &= 0.1096884778767498. \end{aligned}$$

They satisfy $\sum_{j=1}^{6} a_j = 1/2$. Define

$$(b_1, \ldots, b_{12}) = (a_1, a_2, a_3, a_4, a_5, a_6, a_6, a_5, a_4, a_3, a_2, a_1).$$

Starting from $\mathcal{Y}^{(0)}$, the implementation alternates adjoint and direct maps:

$$\mathcal{Y}^{(j)} = \begin{cases} D^*_{b_j h, \tau_{j-1}}(\mathcal{Y}^{(j-1)}), & j \text{ odd}, \\ D_{b_j h, \tau_j}(\mathcal{Y}^{(j-1)}), & j \text{ even}, \end{cases}$$

with

$$\tau_0 = t_n, \qquad \tau_j = t_n + h \sum_{k=1}^{j} b_k.$$

The complete map is

$$\boxed{\Phi^{\mathrm{BM4}}_{h,t_n}(\mathcal{Y}^{(0)}) = \mathcal{Y}^{(12)}.}$$

Negative coefficients are intentional. The direct/adjoint alternation and palindromic coefficients produce the symmetric fourth-order base composition used internally by `BM4Implicit`.

5 Symmetric projection and reduced equation

Given the physical input $\mathcal{Y}_n = Ez_n$, displace it by an unknown multiplier:

$$\widehat{\mathcal{Y}}_n(\mu) = Ez_n + N\mu.$$

Apply the full BM4 cycle and then the same normal correction at the output:

$$\widetilde{\mathcal{Y}}_{n+1}(\mu) = \Phi_{h,t_n}^{\mathrm{BM4}}(Ez_n + N\mu),$$

$$\mathcal{Y}_{n+1}(\mu) = \widetilde{\mathcal{Y}}_{n+1}(\mu) + N\mu.$$

The diagonal condition $G\mathcal{Y}_{n+1} = 0$ gives the reduced equation

$$\boxed{R(\mu) = G\Phi_{h,t_n}^{\mathrm{BM4}}(Ez_n + N\mu) + 2\mu = 0.} \tag{1}$$

If $\Phi_{h,t_n}^{\mathrm{BM4}}(Ez_n + N\mu) = (u_f(\mu), v_f(\mu))$, this is simply

$$u_f(\mu) - v_f(\mu) + 2\mu = 0.$$

At the converged root, the physical output is represented by

$$\boxed{z_{n+1} = \frac{u_f(\mu_*) + \mu_* + v_f(\mu_*) - \mu_*}{2} = \frac{u_f(\mu_*) + v_f(\mu_*)}{2}.} \tag{2}$$

6 BM4 map Jacobian and Newton system

Let

$$J_{\mathrm{BM4}}(\widehat{\mathcal{Y}}_n) = D\Phi_{h,t_n}^{\mathrm{BM4}}(\widehat{\mathcal{Y}}_n).$$

The exact derivative can be accumulated through the twelve stages. If $J_j = D\mathcal{S}_j(\mathcal{Y}^{(j-1)})$ is the Jacobian of stage j, then

$$M_0 = I_4, \qquad M_j = J_j M_{j-1}, \qquad J_{\mathrm{BM4}} = M_{12}.$$

SymPy-explicit stage factors

The following expression makes every factor in J_{BM4} explicit. For $s_j = b_j h$, define the two coupling blocks

$$H_j = \frac{I_2 + R(2\omega s_j)}{2}, \qquad K_j = \frac{I_2 - R(2\omega s_j)}{2},$$

so the Jacobian of the exact linear coupling is

$$C_j = \begin{pmatrix} H_j & K_j \\ K_j & H_j \end{pmatrix}.$$

For an even, direct stage j, let

$$\widetilde{v}_j = v^{(j-1)} + s_j f(u^{(j-1)}, \tau_j),$$

and set

$$U_j = W(u^{(j-1)}, \tau_j), \qquad V_j = W(\widetilde{v}_j, \tau_j), \qquad L_j = I_2 + s_j^2 V_j U_j.$$

SymPy multiplication of the two shears and the final coupling gives the exact 4×4 direct-stage Jacobian

$$\boxed{\begin{aligned}\mathcal{D}_j &= C_j \begin{pmatrix} L_j & s_j V_j \\ s_j U_j & I_2 \end{pmatrix} \\ &= \begin{pmatrix} H_j L_j + s_j K_j U_j & s_j H_j V_j + K_j \\ K_j L_j + s_j H_j U_j & s_j K_j V_j + H_j \end{pmatrix}.\end{aligned}}$$

For an odd, adjoint stage, first write the coupled input as

$$\begin{pmatrix} \overline{u}_j \\ \overline{v}_j \end{pmatrix} = C_j \begin{pmatrix} u^{(j-1)} \\ v^{(j-1)} \end{pmatrix}, \qquad u^{(j)} = \overline{u}_j + s_j f(\overline{v}_j, \tau_{j-1}).$$

Now define

$$U_j = W(\overline{v}_j, \tau_{j-1}), \qquad V_j = W(u^{(j)}, \tau_{j-1}), \qquad L_j = I_2 + s_j^2 V_j U_j.$$

The exact adjoint-stage Jacobian is

$$\boxed{\begin{aligned}\mathcal{A}_j &= \begin{pmatrix} I_2 & s_j U_j \\ s_j V_j & L_j \end{pmatrix} C_j \\ &= \begin{pmatrix} H_j + s_j U_j K_j & K_j + s_j U_j H_j \\ s_j V_j H_j + L_j K_j & s_j V_j K_j + L_j H_j \end{pmatrix}.\end{aligned}}$$

The order of every U_j, V_j, H_j, and K_j factor is essential: these 2×2 matrices do not commute in general.

Let $\mathcal{A}_j[c]$ or $\mathcal{D}_j[c]$ mean that the corresponding stage duration is $s_j = ch$. Substitution of the palindromic BM4 coefficients and application of the chain rule gives the requested explicit expression

$$\boxed{\begin{aligned}J_{\mathrm{BM4}} = \mathcal{D}_{12}[a_1]\mathcal{A}_{11}[a_2]\mathcal{D}_{10}[a_3]\mathcal{A}_9[a_4]\mathcal{D}_8[a_5]\mathcal{A}_7[a_6] \\ \times \mathcal{D}_6[a_6]\mathcal{A}_5[a_5]\mathcal{D}_4[a_4]\mathcal{A}_3[a_3]\mathcal{D}_2[a_2]\mathcal{A}_1[a_1].\end{aligned}}$$

This factorization is exact, not a finite-difference approximation. The reproducible calculation in `bm4_jacobian_sympy.py` uses SymPy 1.14, verifies both displayed stage identities, and preserves noncommutative block order. Even before the internal stage blocks are expanded, the fully expanded twelve-factor product contains 2048 ordered monomials in each final 2×2 block, or 8192 in total. The factorized expression above is therefore the explicit form suitable for analysis and implementation.

Reduced Newton matrix

The reduced Jacobian follows directly from (1):

$$\boxed{K(\mu) = R'(\mu) = G J_{\mathrm{BM4}}(E z_n + N\mu)N + 2I_2.} \tag{3}$$

Newton therefore solves

$$K(\mu^{(k)})\Delta\mu^{(k)} = -R(\mu^{(k)}), \qquad \mu^{(k+1)} = \mu^{(k)} + \Delta\mu^{(k)}.$$

The default implementation computes J_{BM4} from the exact ordered product of the twelve stage factors above. A centered-difference fallback is available for verification. For coordinate j it uses

$$[J_{\mathrm{BM4}}]_{:,j} \approx \frac{\Phi^{\mathrm{BM4}}_{h,t_n}(\widehat{\mathcal{Y}} + \varepsilon_j e_j) - \Phi^{\mathrm{BM4}}_{h,t_n}(\widehat{\mathcal{Y}} - \varepsilon_j e_j)}{2\varepsilon_j},$$

$$\varepsilon_j = \eta \max(1, |\widehat{Y}_j|), \qquad \eta = \sqrt[3]{\epsilon_{\mathrm{mach}}}.$$

No matrix inverse is formed explicitly; the reduced linear system is passed to a dense solve.

7 Stopping rule, symmetry, and order

The initial multiplier is $\mu^{(0)} = 0$. With

$$s_z = \max(1, \|z_n\|_\infty),$$

the solve stops when

$$\left\|R(\mu^{(k)})\right\|_\infty \le \text{atol} + \text{rtol}\, s_z.$$

At an exact, locally unique projection root, the symmetric placement of the same multiplier around the symmetric BM4 map gives

$$\Psi_{-h,t_{n+1}} \circ \Psi_{h,t_n} = I.$$

Under the usual regularity and non-singularity assumptions, symmetric projection preserves the fourth-order accuracy of the BM4 base map. The diagonal defect of a fourth-order cycle is $O(h^5)$, hence the multiplier is expected to satisfy

$$\boxed{\|\mu_h\|_\infty = O(h^5).}$$

If the doubled BM4 map preserves the extended symplectic form and the projection root is solved exactly, Hairer's symmetric projection induces a symplectic map on the physical diagonal. Finite Newton tolerances, finite-difference Jacobians, and floating-point linear solves determine the measured numerical floor.

8 Reproducible numerical study

The following audit uses one guiding-centre particle and the explicit parameters

- random potential amplitude 0.3, maximum wave number 8, grid 32×32, seed 27, interpolation order 5;
- initial state $z_0 = (1, 1.2)$, normalized gyroradius $\rho = 0.3$;
- coupling frequency $\omega = \pi/8$ and interval $[0, 0.4]$;
- Newton absolute and relative tolerances 10^{-15}, at most 12 iterations, and $\eta = \sqrt[3]{\epsilon_{\text{mach}}}$;
- reference solution computed with $h = 0.003125$.

h	global error	observed order	$\max \|\mu\|_\infty$	max Newton it.
0.200	1.2480×10^{-9}	—	6.0004×10^{-10}	1
0.100	7.7277×10^{-11}	4.013	2.1510×10^{-11}	1
0.050	4.8223×10^{-12}	4.002	7.1747×10^{-13}	1
0.025	3.0344×10^{-13}	3.990	2.3095×10^{-14}	1

The multiplier orders between successive rows are 4.802, 4.906, and 4.957, approaching the predicted value five. The maximum final residuals are between 1.00×10^{-15} and 1.83×10^{-15}. These results verify fourth-order convergence and the expected projection scale without using a fitted symbolic model.

9 Implementation correspondence

The production path is:

- `BM4Implicit` selects the reduced solver;

- `_advance_composition` evaluates the unchanged twelve-stage BM4 map;
- `_bm4_map_jacobian` differentiates the complete doubled map;
- `_solve_reduced_projected_bm4_step` assembles (1)–(3);
- `integrate_fixed_grid` advances only physical projected states;
- the reusable BM4 implicit symplecticity study records area, local-flow, accumulated-flow, determinant, residual, iteration, and multiplier data.